Warm Dark Matter Subhalo Suppression in a 2D GPU Cosmology Testbed

Oct 21, 2026 · @John Rice

Note for the expert reader. This is a two-dimensional model universe built as a methods testbed, not a constraint on dark matter. The parts most worth a subhalo specialist's scrutiny are the 2D unbinding criterion (Section 3.2), the unexplained subhalo excess in the M_hm = 300 model (Sections 5.2 and 7.4), the completeness and softening systematics (Section 6), and the list of what does and does not transfer to three dimensions (Section 7.1). The code and this text were written with AI assistance; a second-seed test of the excess has not yet been run.

Abstract

We present a controlled testbed for subhalo statistics: a two-dimensional comoving P³M N-body code running on one consumer GPU, a two-dimensional port of SUBFIND with merger trees, and a pipeline that fits warm-dark-matter (WDM) suppression of the subhalo M_peak function. The fiducial experiment evolves 6144^2 particles on a 4096^2 mesh from a = 0.018 to a = 1 with linear growth exactly D = a, for cold dark matter (CDM) and three WDM models with half-mode masses M_hm = 30, 300 and 3000 particle masses. One model takes 12–18 minutes to simulate and 4 minutes to post-process, so full convergence suites and finder studies fit in an afternoon. The fitted suppression reaches 50 per cent at M_50/M_hm = 0.398 [0.385, 0.416], a factor 6–7 below the three-dimensional values of 2.66 (Lovell et al. 2014) and 2.50 (COZMIC I). Re-expressing masses through σ(M) does not remove the difference, and part of it reflects the geometric convention for M_hm in two dimensions; no amplitude here transfers to three dimensions. The M_hm = 300 model shows an unexplained 10–30 per cent excess of subhaloes over CDM at 1–5 M_hm, which the standard suppression form cannot represent, so the fitted emulator fails closure for that model. A mock satellite limit, M_hm < 479 particle masses, is a pipeline-closure exercise, not a dark-matter constraint. All convergence tests pass at 10 per cent or 2σ except the half-suppression mass under doubled softening (0.80 ± 0.08 of base) and single P(k) bins near the mesh scale or in the fundamental mode of a second seed.

1 Introduction

Counts of dark-matter subhaloes around Milky-Way-like hosts are among the most direct probes of the small-scale matter power spectrum. In warm dark matter (WDM) the free-streaming cutoff suppresses the formation of haloes below the half-mode mass M_{hm} (Viel et al. 2005; Schneider et al. 2012), and the suppression of the subhalo mass function has been calibrated in three-dimensional simulations (Lovell et al. 2014; Lovell 2020; Stücker et al. 2022; Nadler et al. 2025, hereafter COZMIC I) and turned into constraints from the Milky Way satellite census (Nadler et al. 2020, 2021a), strong lensing (Gilman et al. 2020; Enzi et al. 2021; Nadler et al. 2021b) and stellar streams (Banik et al. 2021). Each of these steps rests on choices that are hard to isolate in a large production simulation: the structure finder and its unbinding criterion (Springel et al. 2001; Onions et al. 2012), artificial disruption of subhaloes (van den Bosch & Ogiya 2018), the removal of spurious fragments along filaments in WDM (Wang & White 2007; Lovell et al. 2014), the matching of haloes between models, and the resolution limit below which counts are incomplete.

This paper describes a testbed in which all of those choices can be varied and re-run in minutes. The price is that the universe is two-dimensional: particles move in a periodic plane under a $1/k^2$ Poisson kernel (logarithmic point-mass potential), as in the classic two-dimensional studies of hierarchical clustering (Melott & Shandarin 1989; Beacom et al. 1991; Bagla, Engineer & Padmanabhan 1998). Two-dimensional gravity preserves the qualitative features a methods test needs (hierarchical collapse, filaments, haloes with substructure that loses mass to the host tidal field, merger trees) while reducing the cost of a 3.8×10^7-particle run from hours on a cluster to minutes on a desktop GPU. It changes the quantitative physics: the growth rate, the collapse threshold, the mass-variance slope, the depth of potential wells, the escape criterion, the tidal field and the drag on orbiting substructure are all different, so no number derived here is a prediction for the three-dimensional Universe. We say this in the abstract and again in Section 7.

The paper is organised as follows. Section 2 describes the simulation code and the runs. Section 3 describes the two-dimensional SUBFIND, the merger trees, host matching and the toy galaxy model. Section 4 collects the validation tests (linear growth, power-spectrum cross-check, halo mass function). Section 5 gives the CDM versus WDM results: subhalo counts, suppression fits, the closure test and the mock satellite limit. Section 6 reports the convergence suite and the finder systematics, including a numerical history of the project (bugs found and fixed) because it affects which runs were used. Section 7 discusses what does and does not transfer to three dimensions, and Section 8 concludes.

2 Simulations

2.1 Equations of motion

Lengths are comoving in units where the box side is L = 2000, time is in units of $1/H_0$ with $H_0 = 1$, and the background expands like Einstein-de Sitter, H = H_0 a^(−3/2), dt = a^(1/2) da/H_0. Each particle has comoving position x and canonical momentum p = a^2 dx/dt, and

```
\frac{d\mathbf{x}}{dt} = \frac{\mathbf{p}}{a^2}, \qquad
\frac{d\mathbf{p}}{dt} = -\frac{1}{a}\nabla\phi, \qquad
\nabla^2\phi = C\,\delta, \qquad \delta = \rho/\bar\rho - 1 .
```

Linearising gives $a^2\delta'' + (3/2)a\delta' - (C/H_0^2)\delta = 0$ (primes are d/da). The choice $C = (3/2)H_0^2$ makes the growing mode exactly D(a) = a and the decaying mode $D_- \propto a^{(-3/2)}$, the two-dimensional analogue of $4\pi G\bar{\rho}a^3 = (3/2)H_0^2$ (verified by a unit test on an exact plane wave and by the measured growth, Section 4.1). The code works in canonical momentum so that the integrator factors are analytic.

2.2 Integrator

The code uses the kick-drift-kick leapfrog of Quinn et al. (1997) with merged kicks between steps $a_n \to a_{n+1}$ and geometric half-step midpoints $a_{n\pm1/2}$:

```
\mathbf{p}_{n+1/2} = \mathbf{p}_{n-1/2} + \mathbf{g}(\mathbf{x}_n)\,K(a_{n-
1/2}, a_{n+1/2}), \qquad
\mathbf{x}_{n+1} = \mathbf{x}_n + \mathbf{p}_{n+1/2}\,D(a_n, a_{n+1}),
```

```
K(a_1,a_2) = \int\frac{dt}{a} = \frac{2}{H_0}\left(a_2^{1/2} -
a_1^{1/2}\right), \qquad
D(a_1,a_2) = \int\frac{dt}{a^2} = \frac{2}{H_0}\left(a_1^{-1/2} -
a_2^{-1/2}\right).
```

Before a snapshot the momenta are synchronised with a partial kick $K(a_{n-1/2}, a_n)$. In science mode the global step is set by a CFL-like criterion on the largest comoving acceleration, $\Delta t = \sqrt{(2\eta\varepsilon/A_{max})}$ with $A = |g|/a^3$ and $\eta = 0.025$, converted to $\Delta\ln a = a^{(-3/2)}\,\Delta t\,H_0$, allowed to grow by at most 25 per cent per step, with a floor of $\Delta\ln a = 0.001$ and a ceiling of 0.01; steps are clamped onto the snapshot epochs. The fiducial CDM run took 1720 steps, 372 of them at the floor. Because the step scales as $\sqrt{\varepsilon}$, the doubled-softening variant of Section 6 takes 1250 steps, so that test is partly a step-size test as

well; the halved-η variant (2470 steps) shows that the step size itself is not a systematic. A_max comes from a GPU reduction that is read back asynchronously; the consequence for reproducibility is discussed in Section 6.6.

2.3 Force computation (P³M)

The long-range force is computed on a periodic N × N mesh (Hockney & Eastwood 1988) with cloud-in-cell (CIC) assignment, a spectral Poisson solve φ_k = −C δ_k S(k)/(k² W_CIC²(k)) with the k = 0 mode removed and the Nyquist row and column zeroed, and a spectral gradient a_k = −ikφ_k. The split is Gaussian, S(k) = exp(−k²r_s²) with r_s = 1.25 mesh cells. In real space the mesh then carries a point mass smeared by a Gaussian of variance 2r_s², whose enclosed mass fraction in two dimensions is 1 − exp[−r²/(4r_s²)], so the short-range complement added by direct summation is

```
\mathbf{g}_{\rm sr}(\mathbf{r}) = g_{\rm pp}\,\frac{\mathbf{r}}{r^2 +
\epsilon^2}\,\exp\!\left(-\frac{r^2}{4r_s^2}\right), \qquad r < r_{\rm cut} =
4.5\,r_s, \qquad g_{\rm pp} = \frac{C L^2}{2\pi N_p},
```

with a Plummer-like softening ε. At r_cut the Gaussian factor is exp(−5.06) = 0.6 per cent. Neighbours are found with a chaining mesh whose cells are at least r_cut wide, rebuilt each step by a GPU counting sort. A second pass assigns every particle a deterministic rank inside its cell before the final scatter, so that the order of the pairwise sum, and hence the floating-point force, is independent of thread scheduling (Section 6.6). Late in a run the pair sum takes 0.75-1.7 s per step; it is submitted in chunks of 4096 workgroups with at most two in flight, because longer single submissions triggered the Windows GPU watchdog (timeout detection and recovery, "device lost"). The chunking does not change the arithmetic: forces are bit-identical to a single dispatch.

2.4 Initial conditions and normalisation

Particles start on a 6144² lattice (cell-centred, a multiple of half the mesh so that lattice harmonics alias only onto k = 0 and the Nyquist modes) and are displaced with the Zel'dovich (1970) approximation. The initial density field has fixed amplitude |δ_k| = √(P$_0$(k)/L²) and random phases from a seeded Hermitian white-noise field, so the expectation of the binned power is exactly (a/a_init)²⟨P$_0$⟩ (Angulo & Pontzen 2016; this is what makes the growth test of Section 4.1 clean). The displacement is Ψ_k = ikδ_k/k², deconvolved by the CIC window and interpolated to the lattice; x = q + (a/a_init)Ψ and p = H$_0$ a^(3/2) Ψ/a_init.

The initial spectrum is a cut power law,

$$P_0(k) = A\,k^{n}\exp[-(kR_c)^2], \qquad n = -1.5, \qquad R_c = 0.35 \text{ mesh cells} = 0.1709,$$

multiplied by $T^2(k)$ for WDM (below). $A = 1.5501\times10^{-2}$ and a_init = 0.018262 were set by two conditions: the linear rms density contrast in a two-dimensional top-hat containing 10^6 particles reaches 1.15 at a = 1 when P(k) is integrated from k = 0, and the rms δ on the mesh at a_init is 0.1, so the Zel′dovich start is far from shell crossing. The first condition was applied with the continuum integral; the periodic box has no modes below 2π/L, and the variance actually realised in the box for that top-hat is 0.841. The consequence is that 10^6-particle hosts are 1.5-2σ rarer than intended and the largest hosts are somewhat below 10^6 particles (the largest CDM FoF group has 9.13×10^5 particles). The normalisation was deliberately left unchanged so that the four models differ only in T(k). k_nl(a = 1) = 0.0183 is about six times the fundamental mode 2π/L = 0.00314, so the largest non-linear structures are not well sampled by the box; this limits host statistics, not the methods tests.

2.5 WDM transfer function and half-mode mass

The WDM models use the Viel et al. (2005) fit,

$$T(k) = \left[1 + (\alpha k)^{2\nu}\right]^{-5/\nu}, \qquad \nu = 1.12, \qquad P_{\rm WDM} = T^2 P_{\rm CDM},$$

with the same phases as CDM. The half-mode wavenumber is defined by T(k_hm) = 1/2, k_hm = (2^(ν/5) − 1)^(1/(2ν))/α, λ_hm = 2π/k_hm, and the two-dimensional half-mode mass is taken as the mass of a disc of diameter λ_hm,

$$M_{\rm hm} = \pi\bar\Sigma\left(\frac{\lambda_{\rm hm}}{2}\right)^2 = \pi\bar\Sigma\left(\frac{\pi}{k_{\rm hm}}\right)^2 ,$$

the direct analogue of the three-dimensional sphere of diameter λ_hm. This is a convention, in both dimensions: it fixes the geometric factor that converts a length scale into a mass, and nothing in the dynamics singles it out. Any other reasonable choice (a disc of radius λ_hm, or of radius 1/k_hm) rescales M_hm by a factor of order 4 and rescales every M/M_hm quoted below by the same factor without changing the shape of the suppression. The three models were chosen to have M_hm = 30, 300 and 3000 particle masses, giving k_hm = 3.1231, 0.98761 and 0.31231 (α = 0.1444, 0.4566, 1.4439). With R_c = 0.35 cells the CDM spectrum keeps power to near the mesh Nyquist frequency (k_Nyq = πN/L = 6.43), so all three cutoffs are resolved by the mesh.

2.6 Runs

Table 1 lists the parameters of the fiducial experiment (`exp-corrected`). All masses in this paper are in particle masses, $m_p = L^2/N_p = 0.10596$ in area units where $\Sigma = 1$.

Table 1. Run parameters of the fiducial experiment (identical for the four models).

Parameter	Value
Particles	$6144^2 = 37\,748\,736$
Mesh	4096^2, cell 0.4883
Box	L = 2000 (periodic)
Mean interparticle spacing d	0.3255
Spectrum	$P_0 = A\, k^{(-1.5)} \exp[-(kR_c)^2]$, $A = 1.5501\times10^{-2}$, $R_c = 0.1709$
Normalisation	$\sigma_{tophat}(10^6\, m_p, a=1) = 1.15$ (continuum), 0.841 in box; $\sigma_{mesh}(a_{init}) = 0.1$
$a_{init} \rightarrow a_{final}$	$0.018262 \rightarrow 1$
Growth	D = a (C = 3/2)
Split	Gaussian, $r_s = 1.25$ cells = 0.6104; $r_{cut} = 4.5\, r_s = 2.747$
Softening	$\varepsilon = 0.026042 = 0.08$ d (Plummer-like)
Time step	CFL $\eta = 0.025$; $10^{-3} \leqslant \Delta\ln a \leqslant 10^{-2}$; growth ⩽ 25 per cent per step
Snapshots	16, log-spaced in a from 0.1 to 1
Seed	1
WDM	$\nu = 1.12$; M_{hm} = 30, 300, 3000 m_p (k_{hm} = 3.123, 0.988, 0.312)
Hardware	one RTX 5080; 12-18 min per model

Wall-clock times per model were 13.8 min (CDM, 1720 steps), 17.8 min (M_{hm} = 3000, 1712 steps), 15.6 min (M_{hm} = 300, 1715 steps) and 11.5 min (M_{hm} = 30, 1715 steps). The fiducial experiment is assembled from two batches: `cdm` and `wdm_khm0.31` are the `control` runs

of the convergence suite, re-run with the current code, while `wdm_khm0.99` and `wdm_khm3.12` are the original runs of experiment `exp-20261021-0022`; the reason, and the measured agreement between the re-runs and the originals, are given in Section 6.7. All four have identical settings, seed and normalisation, and all four end with a mean velocity of $3\text{-}5\times10^{-5}$ against an rms of 220.

3 Structure finding and galaxy model

3.1 Friends-of-friends groups and hosts

Groups are found with friends-of-friends (FoF; Davis et al. 1985) at linking length b = 0.2 times the mean interparticle spacing in every snapshot, keeping groups of at least 20 particles. Every particle's index in the snapshot is its stable ID (its lattice index), which the finder, the trees and the host matching rely on. A host radius R_host is defined for each group as the largest radius around the central subhalo's most-bound particle within which the mean enclosed surface density of FoF members is at least 50 times the cosmic mean. Hosts used in the analysis have at least 1000 FoF particles.

3.2 Two-dimensional SUBFIND

The finder follows Springel et al. (2001) step by step, applied to each FoF group:

1. Densities are SPH estimates from the k_dens = 32 nearest group members with the two-dimensional cubic spline $W = (40/(7\pi h^2))\{1 - 6q^2 + 6q^3\ (q < 1/2);\ 2(1-q)^3\ (q < 1)\}$.
2. Particles are processed in order of decreasing density. The two nearest already-processed (denser) particles among the n_ngb = 20 neighbours decide: none, a new local maximum; one or both in the same subgroup, join it; two different subgroups, the particle is a saddle point, the smaller subgroup is recorded as a candidate if it has at least n_min = 20 members and the two are merged. Candidates are nested.
3. Candidates are unbound in order of increasing size; each keeps only particles not already claimed by a smaller accepted subhalo. Each iteration recomputes the potential (a kd-tree Barnes-Hut sum of the softened logarithmic kernel with opening angle $\theta = 0.5$, or the direct sum for small sets), the potential minimum and the mean velocity, and removes the particles with $E > 0$, at most 25 per cent of the set per iteration, most unbound first. A candidate with at least n_min bound particles is accepted.
4. The largest accepted subhalo of a group is its central; the others are satellites.

The binding criterion needs a two-dimensional treatment. With the peculiar velocity v = a dx/dt = p/a and physical coordinates r = ax, E = v²/2 + Φ with Φ = φ/a is the peculiar energy per unit mass, conserved apart from Hubble drag (ignored, as in SUBFIND). In area units (m = L²/N_p, Σ = 1) the Green's function of the Laplacian is ln|x|/2π and ∇²|x|² = 4, so the full peculiar potential of a particle set, including the uniform background term from δ = ρ/ρ̄ − 1, is

```
\phi(\mathbf{x}) = \frac{Cm}{2\pi}\sum_j \ln\sqrt{|\mathbf{x} -
\mathbf{x}_j|^2 + \epsilon^2} - \frac{C}{4}|\mathbf{x} - \mathbf{x}_c|^2 +
\text{const}.
```

The logarithmic potential has no zero at infinity, so "E < 0 with Φ(∞) = 0" has no analogue: an isolated two-dimensional mass binds every particle. "Bound" therefore has to be defined relative to a reference level, and the choice is a convention of this finder rather than a property of the physics. Energies are measured relative to the candidate's own potential at a reference radius R_ref = f_esc R_ext, where R_ext is the extent of the initial candidate (distance of its farthest member from the potential minimum, i.e. the SUBFIND saddle-point boundary), frozen for the iteration:

```
E_i = \tfrac{1}{2}|\mathbf{v}_i - \bar{\mathbf{v}}|^2
 + \frac{C}{2\pi a}\Big[m\sum_{j \ne i}\ln\sqrt{r_{ij}^2 + \epsilon^2} - M\ln
R_{\rm ref}\Big]
 - \frac{C}{4a}\big(|\mathbf{x}_i - \mathbf{x}_c|^2 - R_{\rm ref}^2\big),
\qquad \text{bound} \iff E_i < 0 .
```

The default f_esc = e reproduces the three-dimensional escape budget at the boundary: in three dimensions getting from the edge R to infinity costs GM/R = v_c²(R); in two dimensions getting from R to fR in a point-mass log potential costs v_c² ln f, so ln f = 1. With f_esc = 1 ("must not reach the saddle") large haloes lose most of their mass in a removal cascade, because the depth of a logarithmic well scales with the remaining mass. The sensitivity to f_esc is measured in Section 6.4 and is carried as a systematic. The softening in the potential is the simulation's PP softening ε = 0.026042 (this was not always so; Section 6.7).

Circular velocities need the background too. Gauss's law gives |∇_x φ| = C(M(<r) − πr²)/(2πr) and

```
v_c^2(r) = \frac{C\,[M(<r) - \pi\bar\Sigma r^2]}{2\pi a} .
```

Without the subtraction v_c would never decrease (a point mass has a flat rotation curve in two dimensions); V_max and R_max exist only because of it, and R_max is where the azimuthally averaged surface density of the subhalo falls to the cosmic mean. V_peak in the galaxy model (Section 3.6) is therefore a background-subtracted quantity with no direct three-dimensional counterpart.

The finder uses 16 threads and processed the 16 CDM snapshots in 223 s; the a = 1 CDM catalogue has 100 225 FoF groups, 56 832 centrals and 28 009 satellites (84 841 subhaloes).

3.3 Merger trees, M_peak and infall

Each subhalo carries its n_tr = 20 most-bound particle IDs weighted 1/(k+1) by binding rank. In the next snapshot every tracer votes for the subhalo containing it, or for the central of its FoF group if it is in no subhalo; the descendant is the candidate with the largest weighted vote provided it holds at least 10 per cent of the tracer weight. A satellite whose tracers land only in a host central or nowhere may be temporarily unresolved near pericentre, so the link is kept pending and the snapshot after next is searched (a one-snapshot skip). Main progenitors are the progenitors with the most bound particles, and following main-progenitor links gives disjoint branches, each ending as surviving (present at a = 1), merged/disrupted, or lost. For the CDM run 822 046 links were searched: 666 243 direct, 31 621 skips, 63 680 into a central, 60 502 lost (linked fraction 0.926). Branches supply M_peak (maximum bound mass along the branch), V_peak, a_infall and a surviving/disrupted flag; there are 273 275 CDM branches of which 84 841 survive and 129 527 are disrupted. The snapshot spacing (16 epochs from a = 0.1, 0.153 in ln a) and the 7 per cent of lost links are a known limitation: a subhalo that peaks between snapshots has its M_peak underestimated by the growth over that interval, and the bias is model-independent only to the extent that the growth histories are.

3.4 Host matching between models

Hosts in different models are matched by particle IDs: FoF (b = 0.2) at a = 1 in both models, the best partner of each group by shared particles, accepted when the match is mutual and the shared fraction is at least 0.5. A purely positional fallback (nearest host within 0.5R_host and a factor 3 in mass) exists but was not used for the results here (Section 6.7). Of the 1792 CDM hosts with at least 1000 particles, 1578 (88.1 per cent), 1400 (78.1 per cent) and 830 (46.3 per cent) are paired in the M_hm = 30, 300 and 3000 models, with median shared fractions 0.89, 0.80 and 0.60. Milky Way analogues are the 30 most massive CDM hosts, $M_{FoF} \in [4.71\times10^{4}, 9.13\times10^{5}]\ m_p$; 28, 28 and 24 of them have counterparts,

and 24 are paired in every model. The paired WDM counterparts are not the same objects at the same mass: their FoF masses exceed those of their CDM partners by a median factor 1.01, 1.05 and 1.14 (16-84 per cent ranges 0.87-1.12, 1.01-1.15, 1.03-1.40) and their R_host by 1, 2 and 4 per cent (Milky Way analogues only). Mass that in CDM sits in small haloes near the host is, in WDM, part of the host. This matters for Section 5.

3.5 Completeness, satellites and cleaning

Satellites of a host are the surviving branches whose a = 1 position lies within R_host (projected distance), excluding the host's own central and branches that were never satellites. Counts and fits use M_peak ⩾ M_complete = 160 m_p, set on the command line from the down-sampling study of Section 6.3. Artificial fragmentation is handled with a two-dimensional transcription of the Wang & White (2007) limit: with k_peak the wavenumber where $k^2P(k)$ peaks and $\kappa = 4 \times 10.1/\pi^3$ carried over from the three-dimensional calibration, $M_{lim,2D} = \kappa \Sigma d \pi/k_{peak} = 4.09 \Sigma d/k_{peak}$. The coefficient is a transcription, not a two-dimensional calibration; the quantity that is actually tested is whether anything near M_lim enters the fit sample. The values are M_lim = 4.30, 10.2, 29.7 and 93.2 m_p for CDM and the three WDM models ($M_{lim}/M_{hm} = 0.34, 0.099, 0.031$), all below M_complete. Following Lovell et al. (2014), a WDM (sub)halo is flagged spurious if its mass is below 0.5M_lim and the sphericity of its Lagrangian proto-halo (axis ratio of the particle distribution at a_init) is below the first percentile of CDM objects with more than 100 particles, s_cut = 0.0328. Flagged objects: 0, 0 and 27 subhaloes (20 of them removed from host satellite lists) in the three WDM models; the fraction of resolved subhaloes with s < s_cut is 0.004 (CDM), 0.003, 0.002 and 0.001. The fit sample above M_complete is unaffected by cleaning.

3.6 Toy galaxy model and likelihood

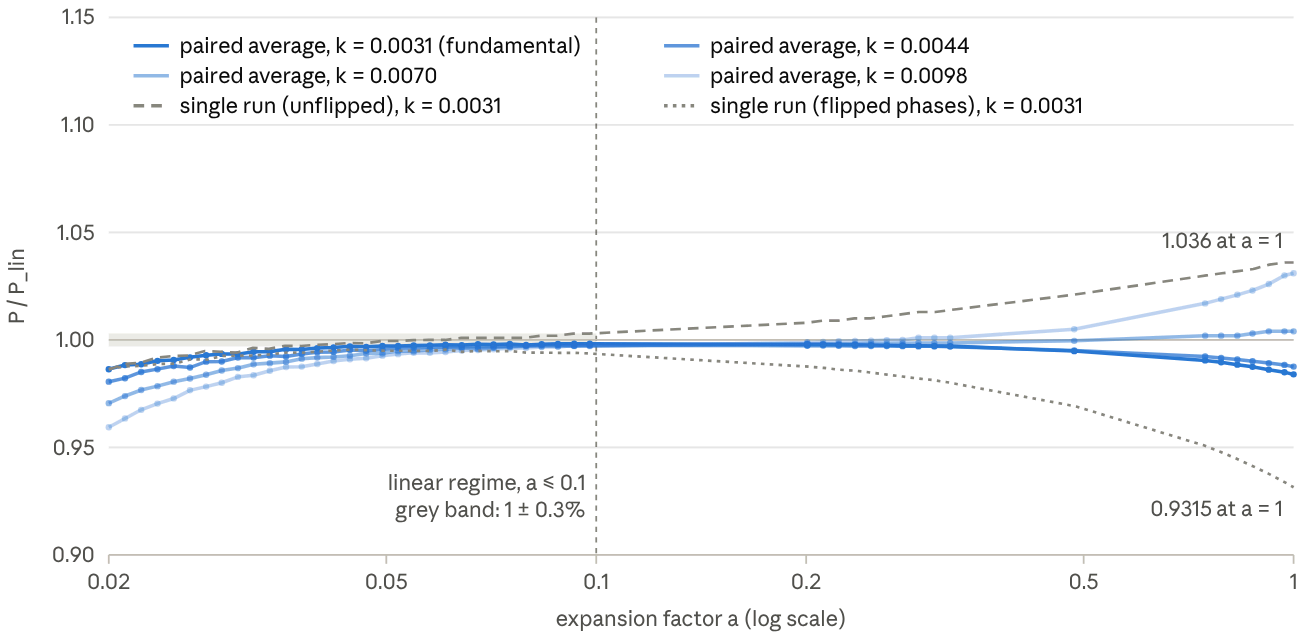
The galaxy model is modelled on the structure of Nadler et al. (2020) and is a toy by construction: none of its parameters is calibrated to anything, and its only purpose is to turn subhalo catalogues into "observable satellite" counts that a likelihood can be run on. A satellite with peak mass M_peak gets $\log_{10}L = 2\log_{10}(M_{peak}/M_{norm}) + \mathcal{N}(0, \sigma_M)$ with M_norm = 100 m_p (a steep slope of the kind used for faint dwarfs). A galaxy forms with probability $f_{gal}(V_{peak}) = \tfrac{1}{2}[1 + \mathrm{erf}((V_{peak} - V_{50})/\sqrt{2}\sigma_{gal})]$, where V_50 is in units of V_res = 2.174, the median V_peak of CDM branches with $80 \leqslant M_{peak}/m_p \leqslant 125$. Disrupted branches keep an orphan with probability $p_{orph} \exp[-(1 - a_{infall})/\tau]$. Fiducial parameters are $V_{50}/V_{res} = 1.5$, $\sigma_{gal}/V_{50} = 0.15$, $\sigma_M = 0.2$ dex, $p_{orph} = 0.5$, $\tau = 0.5$. The observability cut $\log_{10}L_{cut} = 0.564$ is tuned so that the fiducial model gives 30 observable satellites per CDM Milky Way analogue; it corresponds to M_peak = 192 m_p,

above M_complete, so the target was reachable and the satellites are "observable" rather than merely resolved. The fiducial V_50 corresponds to M_peak = 239 m_p, so the model is threshold-limited. Luminosity-function bins are 0.5 dex wide from L_cut with an open last bin (8 bins). Expected counts are sums of per-branch probabilities; a mock survey draws Bernoulli occupation and lognormal luminosities. The likelihood is Poisson per bin, marginalised over 1000 draws of the galaxy parameters from either a fiducial prior ($V_{50}/V_{res} \in [1.2, 1.9]$ log-uniform, $\sigma_{gal} \in [0, 0.3]$, $\sigma_M \in [0.1, 0.3]$, $p_{orph} \in [0, 1]$) or a wide prior ($V_{50}/V_{res} \in [0.5, 4]$).

4 Validation

4.1 Linear growth

A pair of PM-only validation runs (directories `20261020-233221` and `20261020-233559`: $n = -1$, 2048^2 mesh, $a_{init} = 0.02$, same seed, the second with all phases flipped, $\delta \to -\delta$) tests $D = a$. The measured power in each k bin is divided by the exact expectation for fixed-amplitude initial conditions, $(a/a_{init})^2 \langle P_0(k) \rangle$. Averaging the two runs cancels the leading realisation-specific term (Angulo & Pontzen 2016). At $a = 0.02$ the paired average in the four lowest-k bins is 0.986, 0.981, 0.971 and 0.959 (the lattice initial conditions are sub-Poisson, so subtracting Poisson shot noise biases the first measurement low); by $a = 0.098$, i.e. $D/D_{init} = 4.9$ and still linear (k_{nl} is undefined before $a = 0.333$), it is 0.998, 0.998, 0.997 and 0.997. The growth factor is therefore reproduced to 0.3 per cent in the linear regime (Fig. 4). The single runs then drift apart (1.036 and 0.932 in the fundamental mode at $a = 1$) through mode coupling once $k_{nl}(a = 1) = 0.061$ is only a factor 20 above the fundamental mode; this is physics, not an integrator error. The paired average at $a = 1$ is 0.984 in the fundamental mode and 1.031 at $k = 0.0098$, i.e. within 3 per cent of linear.



figures_v2.json · pk.csv of runs 20261020-233221 and 20261020-233559

4.2 Power spectrum: GPU versus offline measurement

The GPU measures P(k) in 40 logarithmic bins every 0.05 in ln a with CIC deconvolution and shot-noise subtraction. An independent CPU measurement from the snapshots (2048² grid) agrees with the GPU values to a median ratio of 1.0005 at every one of the six snapshot epochs of run `20261020-233820` (a = 0.1 to 1; mean |ln ratio| 0.004-0.011 over 27-34 common bins). The two codes share no measurement code.

4.3 Halo mass function in two dimensions

The same run was used to compare the FoF halo mass function with excursion-set forms. The two-dimensional circular top-hat collapse threshold computed by the code is δ_c = 1.4662 (the same routine gives 1.6865 for the three-dimensional sphere, the familiar 1.686). The multiplicity functions of Press & Schechter (1974) and Sheth & Tormen (1999) are dimension independent in form; only σ(M) (two-dimensional top-hat, M = ΣπR²), p̄ per unit area and δ_c change. At a = 1 the FoF catalogue has 34 808 groups of at least 20 particles holding 31 per cent of the mass, 9801 with at least 100 particles. Poisson (Cash) statistics over 34 mass bins: Press-Schechter with the theoretical δ_c, 9131; Press-Schechter with δ_c free, 591 at an unphysical δ_c = 0.48; Sheth-Tormen with (p, q) fitted, 564 at p = 0.38, q = 0.25 (δ_c√q = 0.73). The two-dimensional FoF mass function is thus far from Press-

Schechter with the top-hat threshold and is described by a Sheth-Tormen form only with fitted parameters. We have not investigated why (FoF at b = 0.2 in two dimensions is itself uncalibrated against any overdensity definition) and use the comparison only as a check that the σ(M) machinery and hierarchical growth behave sensibly, not as a result.

4.4 Force and determinism tests

A GPU two-point force test (mesh only versus P³M) is built into the code, and a CPU mirror of the mesh force (CIC deposit, Gaussian split, W^2 deconvolution) is used in unit tests. The P³M determinism test (Section 6.6) ran the full 6144^2 system for 100 fixed steps twice: 0 particles differed.

5 Results

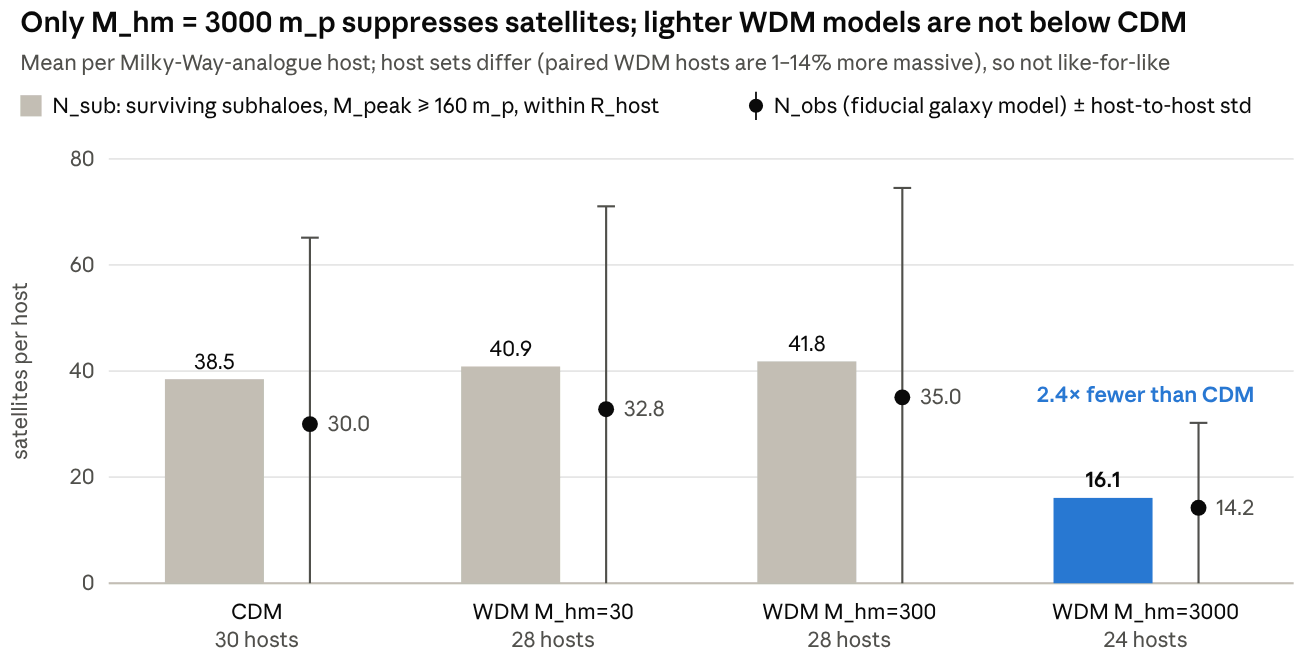
5.1 Catalogues and counts per host

Table 2 gives the per-model numbers. The two lighter WDM models have slightly more subhaloes per Milky Way analogue than CDM (40.9 and 41.8 against 38.5), not fewer; only M_hm = 3000 m_p shows net suppression, by a factor 2.4. Two cautions apply to this comparison. The CDM mean is over 30 hosts and the WDM means over the 28, 28 and 24 paired hosts, so the host sets differ; and the paired WDM hosts are 1-14 per cent more massive than their CDM partners (Section 3.4), which by itself raises their subhalo counts. The like-for-like comparison is the pair-level ratio of Section 5.2 and the closure test of Section 5.4. The host-to-host standard deviation of the observable count is comparable to the mean (35, 38, 39, 16), because the top-30 hosts span a factor 20 in mass.

Table 2. Per-model numbers of the fiducial experiment. Hosts: FoF groups with ⩾ 1000 particles. N_sub: surviving subhaloes with M_peak ⩾ 160 m_p inside R_host per Milky Way analogue (30 CDM hosts; 28, 28, 24 paired WDM hosts). N_obs: fiducial-model observable satellites per analogue, with the host-to-host standard deviation.

Model	M_hm/m_p	k_hm	Hosts	Subhaloes (a=1)	Branches	Paired hosts (per cent)	MW pairs	N_sub	N_obs ± std	M_lim/m_p	Spurious flagged
CDM	0	0	1792	84 841	214 368	100	30	38.5	30.00 ± 35.14	4.30	0
WDM	30	3.123	1805	66 875	162 379	88.1	28	40.9	32.81 ± 38.24	10.2	0

Model	M_hm/m_p	k_hm	Hosts	Subhaloes (a=1)	Branches	Paired hosts (per cent)	MW pairs	N_sub	N_obs ± std	M_lim/m_p	Spurious flagged
WDM	300	0.988	1668	25 000	49 424	78.1	28	41.8	35.04 ± 39.47	29.7	0
WDM	3000	0.312	1218	5 486	14 128	46.3	24	16.1	14.22 ± 16.01	93.2	27



figures_v2.json · experiment_summary.json (satellites_per_model)

The projected radial distribution of satellites in Milky Way analogues rises from 2.4 subhaloes in the innermost annulus of width 0.1R_host to 5.6-6.2 at 0.2-0.4R_host, stays at 4.7-5.6 out to 0.8R_host and falls to 3.5-3.9 in the outer two annuli (CDM, 30 hosts); the M_hm = 3000 model has a similar shape at 1.2-2.2 per annulus. We draw no conclusion about tidal disruption from this: the inner deficit is in projection, in a logarithmic host potential, and at a resolution where a = 1 bound masses near the centre can be far below M_peak.

5.2 Suppression of the subhalo M_peak function

For each WDM model, subhaloes of all paired hosts are counted in bins of M_peak (0.2 dex) and compared with the CDM counts in the paired CDM hosts. The suppression function

```
f(M) = \left[1 + \left(\alpha\,\frac{M_{\rm hm}}
{M}\right)^{\beta}\right]^{\gamma}
```

is fitted jointly to the three models with a conditional-binomial likelihood (N_WDM | N_tot ~ Bin(N_tot, f/(1+f)), which profiles out the CDM expectation per bin), with f averaged over each bin with a M^(−0.9) weight, using bins with M_peak ⩾ 160 m_p, parameters (ln α, ln β, ln(−γ)) clamped to [−8, 8], [−3, 5], [−5, 2.5], 27 Nelder-Mead starts, and 200 pair-level bootstrap resamples for the 16-84 per cent ranges. Table 3 gives the result together with the halo (central) fit and the three-dimensional reference fits. The best fit is not at any clamp; 6 per cent of bootstrap samples hit one.

Table 3. Suppression fits. Ranges are bootstrap 16-84 percentiles. M_50 is the mass at which f = 1/2.

Fit	α	β	γ	M_50/M_hm
2D subhaloes, M_peak (this work)	0.856 [0.811, 0.913]	8.33 [6.18, 14.70]	−0.109 [−0.146, −0.061]	0.398 [0.385, 0.416]
2D central haloes (this work)	0.085 [0.049, 0.119]	1.12 [1.07, 1.18]	−3.83 [−6.09, −2.89]	0.360 [0.347, 0.370]
3D Lovell et al. 2014, genuine subhaloes	2.7	1	−0.99	2.66
3D COZMIC I (Nadler et al. 2025), M_peak	2.5	0.9	−1	2.50
3D Lovell 2020, subhalo bound mass	4.2	2.5	−0.2	1.06
3D Lovell 2020, centrals	2.3	0.8	−1	2.30

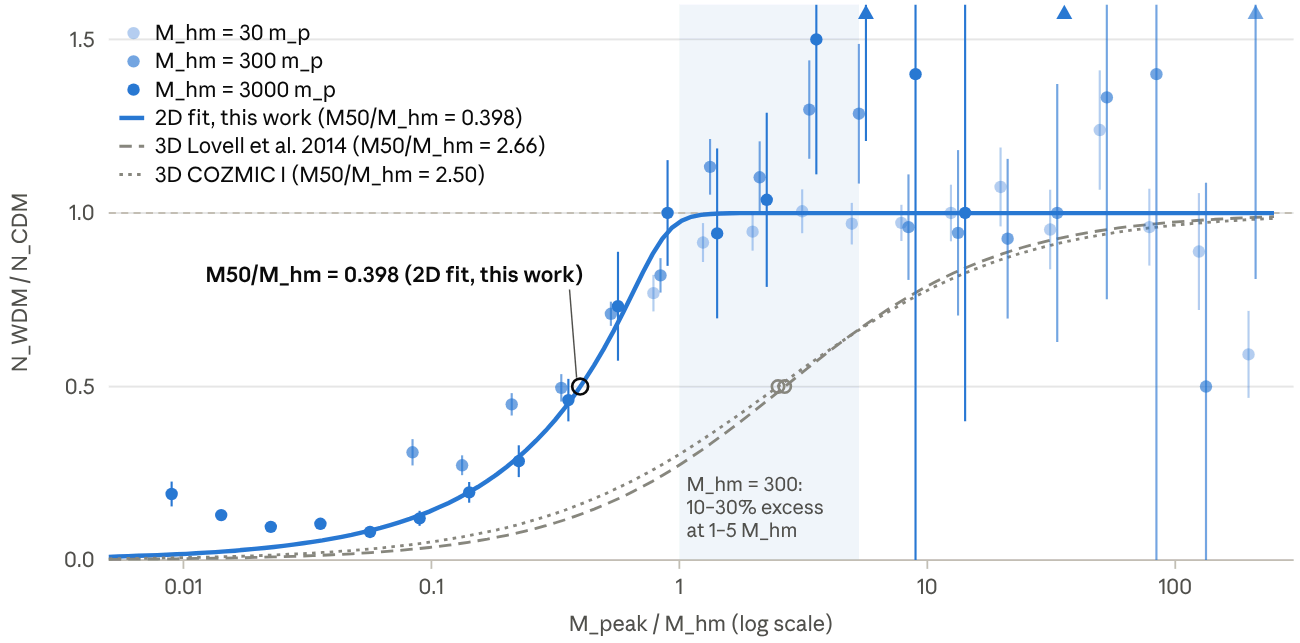
The two-dimensional suppression is much sharper and sets in much lower: f(M_hm) = 0.97, f(M_hm/3) = 0.43, f(M_hm/30) = 0.053, whereas the three-dimensional fits give f(M_hm) = 0.27 (Lovell et al. 2014) and 0.31 (COZMIC I) and reach only 0.55 at 3M_hm. The ratio of half-suppression masses is 6.7 (Lovell et al. 2014) and 6.3 (COZMIC I); as noted in Section 2.5, a factor of order 4 of this is at the mercy of the geometric convention for M_hm. The three-parameter form is strongly degenerate here (β and γ trade off, which is why their ranges are wide), and M_50 is the robust summary.

The per-bin ratios for the Milky Way analogues are plotted in Fig. 1. For M_hm = 300 they are 0.71 ± 0.03 at M_peak/M_hm = 0.53 and 0.82 ± 0.05 at 0.84, then 1.13 ± 0.08, 1.10 ± 0.10, 1.30 ± 0.14 and 1.29 ± 0.20 at 1.3, 2.1, 3.3 and 5.3, and consistent with unity above that. Four consecutive bins above unity by 10-30 per cent, each at 1-2σ, is an excess that the fit form (which approaches unity from below) cannot represent, and the fit ignores it. For M_hm = 3000 the same bins are 0.94 ± 0.24, 1.04 ± 0.25, 1.5 ± 0.4 and 2.0 ± 0.8, suggestive but with few objects. For M_hm = 30 the range 1-5 M_hm is 30-150 m_p, below M_complete, and the bins above M_complete (which start at 5 M_hm) scatter about unity (0.97, 1.00, 1.07, 0.95, 1.24, 0.96, 0.89, 0.59, ±0.05-0.17); its 6 per cent higher per-host count in Table 2 is a mixture of these bins and of the host-set differences already mentioned. The excess is unexplained; Section 7.4 lists the candidate causes and the tests that would separate them.

2D suppression halves at M_peak ≈ 0.40 M_hm, 6–7× lower than in 3D

WDM/CDM surviving subhaloes per Milky-Way-analogue host, ±1σ pair bootstrap; 3D fits grey dashed

▲ = ratio above 1.6, drawn at the top edge (hover for value)



figures_v2.json · exp-corrected/analysis/shmf.csv

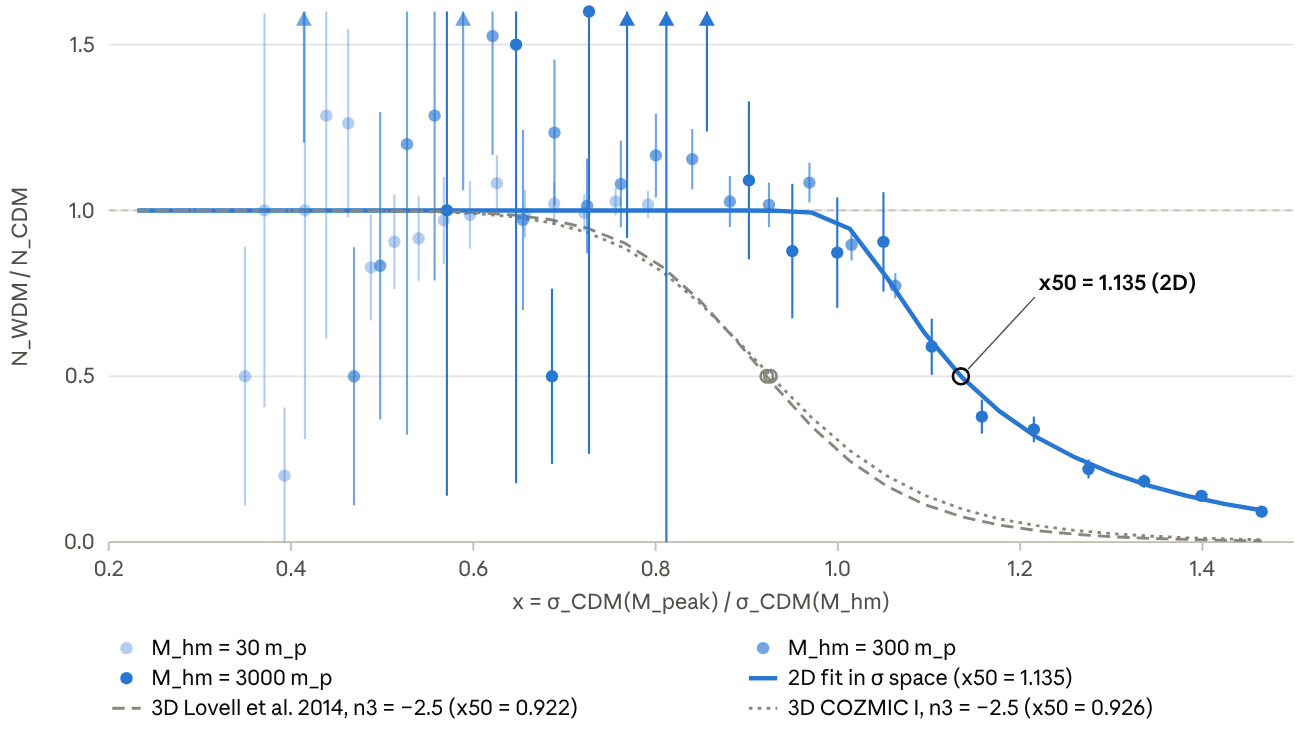
5.3 Suppression in σ space

Because the mass-variance slope differs between dimensions, one might hope that the suppression is universal in x = σ_CDM(M)/σ_CDM(M_hm) (in-box two-dimensional top-hat on the CDM linear spectrum). The same bin ratios refitted in x give α = 0.980 [0.971, 0.988], β = 58.2 [41.1, 118.1], γ = −0.111 [−0.157, −0.052] and x_50 = 1.135 [1.130, 1.142] (the

printed "M_50/M_hm" column of the sigma fit is 1/x_50 = 0.881; 14 per cent of bootstrap samples hit a clamp, reflecting how step-like the function is in x). Mapping the three-dimensional fits into x with σ_3D ∝ M^(−(n_3+3)/6) and n_3 = n_2D − 1 = −2.5 (the correspondence that gives the same σ(R) slope in both dimensions, as in Beacom et al. 1991), so that M_hm/M = x¹², gives x_50 = 0.922 (Lovell et al. 2014), 0.926 (COZMIC I) and 0.995 (Lovell 2020 bound mass). The measured range is x ∈ [0.23, 1.47], so the comparison is inside the data. There is no agreement in σ space either (Fig. 5).

In σ space 2D still halves at x50 = 1.135, against 0.92–0.93 in 3D

WDM/CDM subhalo ratio (M_peak ⩾ 160 m_p), ±1σ; larger x = smaller mass; ▲ = ratio above 1.6 at the top edge



figures_v2.json · suppression_sigma.csv

5.4 Closure test

The fitted f(M_peak) is an emulator: applied to the CDM satellites of a paired host it should reproduce the number of observable satellites in the WDM counterpart. For the fiducial galaxy model on the Milky Way analogues (Table 4), the emulator passes within 2√N_direct for M_hm = 30 and 3000 but fails for M_hm = 300, where it underpredicts by 12 per cent. The emulator is low for all three models (by 3, 12 and 5 per cent), which is the signature of an excess over CDM that a form bounded by f ⩽ 1 cannot reproduce. The

host-to-host bootstrap error of the direct sum (also listed) is much larger than the counting error, so the test as defined is strict. Because closure is not met, the analysis does not report an emulator-based limit.

Table 4. Closure of the emulator. Pass if |N_emu − N_direct| ⩽ 2√N_direct.

Model	Hosts	N_direct	√N	Host bootstrap	N_emu	Ratio	Pass
M_hm = 30	28	918.6	30.3	202.2	889.3	0.968	yes
M_hm = 300	28	981.0	31.3	200.7	865.3	0.882	no
M_hm = 3000	24	341.2	18.5	78.7	322.7	0.946	yes

5.5 Mock survey and the pipeline-closure limit on M_hm

The mock "observation" is one draw of the fiducial galaxy model on the 24 CDM hosts paired in every model: 826 satellites in the eight luminosity bins, [207, 220, 156, 96, 61, 44, 17, 25]. The "direct" likelihood uses each WDM simulation's own paired satellites, with ln μ interpolated linearly in ln M_hm between the simulated nodes, CDM anchored at M_hm = 3 (a tenth of the smallest simulated value) and no extrapolation above 3000. Because of the WDM excesses, a limit that rewards WDM for *more* satellites is not conservative, so a capped variant sets each WDM prediction to at most the CDM prediction per bin and per parameter draw, and a further variant marginalises a lognormal 10 per cent satellite-count nuisance representing the finder's f_esc systematic (Section 6.4). Table 5 lists 95 per cent upper limits from the profile likelihood ratio (Δln L = 1.353) and from the posterior with a prior flat in log M_hm over the grid (the posterior limit depends on the grid's lower edge and is quoted only for completeness). The uncapped fiducial-prior LR limit (28.8) lies below the smallest simulated M_hm and is set by interpolation towards the CDM anchor; it is anchor-dependent and should be read as "< 30".

Table 5. 95 per cent upper limits on M_hm (particle masses) from the mock survey.

Method	Fiducial prior, LR	Fiducial, posterior	Wide prior, LR	Wide, posterior
Emulator (fitted f)	not reported (fails closure)	-	-	-
Direct interpolation	28.8 (below lowest node)	401.7	534.8	477.3

Method	Fiducial prior, LR	Fiducial, posterior	Wide prior, LR	Wide, posterior
Direct, capped at CDM	419.6	343.8	441.6	367.4
Direct, capped, + 10 per cent count systematic	478.9	389.8	516.5	420.1

The limit we quote is the capped, systematic-marginalised profile-likelihood one, M_hm < 479 m_p (fiducial prior) and < 517 m_p (wide prior). What it tests is the pipeline, not dark matter: the mock data are a CDM realisation of the same galaxy model, the "truth" is M_hm = 0, the universe is two-dimensional, and the result says that with 826 satellites in 24 hosts the likelihood machinery excludes the M_hm = 3000 model and not the M_hm = 300 one, with the limit interpolated between those two nodes. It is sensitive to the interpolation scheme (three nodes a decade apart), to the galaxy-model prior (fiducial versus wide shifts it by 8 per cent) and to the capping. We call it a closure exercise throughout.

6 Convergence and systematics

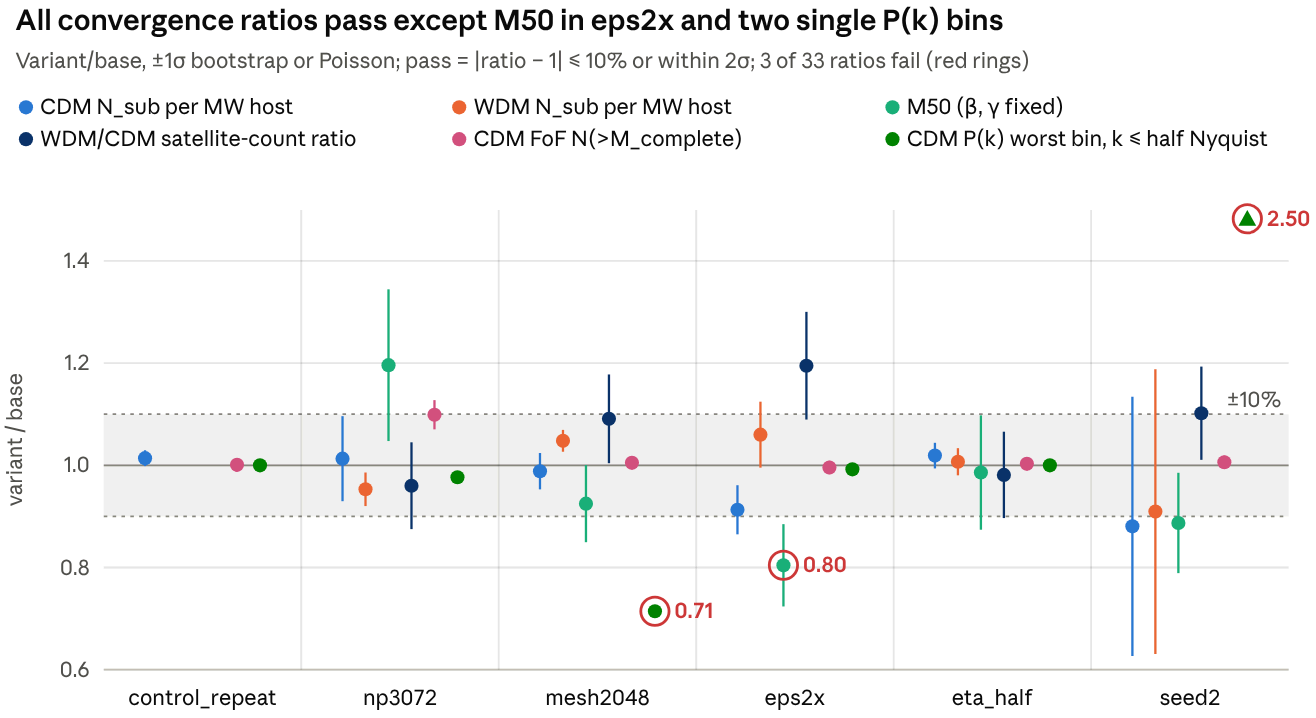
6.1 Convergence suite

Six variants of the base experiment were run, each changing one thing (`conv-20261021`): `control_repeat` (identical settings and seed, CDM only), `np3072` (3072^2 particles, particle mass ×4, same initial field sampled coarser), `mesh2048` (PM mesh 2048^2, so r_s, r_cut and the PP range double), `eps2x` (ε = 0.052), `eta_half` (η = 0.0125, 2470 steps) and `seed2` (different phases; cosmic variance, not convergence). Each variant has a CDM and an M_hm = 3000 run except `control_repeat`. Hosts are matched to the base through Lagrangian particle overlap (a coarser lattice particle covers r × r base cells), mutual best match with shared fraction ⩾ 0.5; `seed2` cannot be matched and its own top-30 hosts are compared statistically. All quantities are compared in physical units at a = 1: for `np3072` the completeness cut is 160 of its own particles, i.e. 640 base particle masses, applied to both runs. The suppression is summarised by M_50 from a single-model fit with β and γ held at the base values (8.33, −0.109); the base value is M_50 = 120.5 in area units = 1137 m_p = 0.38 M_hm. A quantity is converged if |ratio − 1| ⩽ 10 per cent or within 2σ (bootstrap or Poisson); P(k) ratios of same-seed variants have no cosmic-variance error, so only the 10 per cent criterion applies to them.

Table 6. Convergence ratios variant/base. N_sub: surviving subhaloes per Milky Way analogue above M_complete; N_FoF: FoF groups above M_complete; P(k) worst: largest single-bin deviation for k ⩽ half Nyquist. Bold entries fail the criterion.

Variant	CDM N_sub	WDM N_sub	M_50	WDM/CDM count ratio	CDM N_FoF	WDM N_FoF	CDM P(k) worst	WDM P(k) worst
control_repeat	1.014 ± 0.016	-	-	-	1.001 ± 0.013	-	1.00003	-
np3072	1.013 ± 0.083	0.953 ± 0.033	1.196 ± 0.149	0.960 ± 0.085	1.099 ± 0.029	1.046 ± 0.036	0.977	0.968
mesh2048	0.989 ± 0.036	1.048 ± 0.021	0.925 ± 0.075	1.091 ± 0.087	1.005 ± 0.013	1.000 ± 0.022	**0.714**	**0.714**
eps2x	0.913 ± 0.048	1.060 ± 0.065	**0.804 ± 0.081**	1.195 ± 0.105	0.996 ± 0.013	0.998 ± 0.022	0.992	0.995
eta_half	1.019 ± 0.025	1.007 ± 0.027	0.986 ± 0.112	0.981 ± 0.085	1.003 ± 0.013	1.011 ± 0.022	1.0003	1.0003
seed2	0.881 ± 0.254	0.909 ± 0.279	0.887 ± 0.098	1.102 ± 0.091	1.006 ± 0.013	0.994 ± 0.022	**2.497**	**2.497**

Everything passes except M_50 under doubled softening and the worst single P(k) bin for `mesh2048` and `seed2`. The `mesh2048` outlier is a single bin near the half-Nyquist edge of the comparison range where the coarser mesh's split moves power between mesh and PP parts (the median ratio is 0.9997); the `seed2` outlier is the fundamental mode of a different realisation (median 1.005) and is cosmic variance. Two passes are marginal and should be said to be: the `np3072` CDM FoF count above 640 base particle masses is 1.099 ± 0.029, 10 per cent high at 3.4σ, so the FoF count at 160 particles is not resolution-independent at the few-per-cent level; and the `np3072` M_50 ratio of 1.20 ± 0.15 is the second-largest shift in the table. With four times the particle mass the resolution test is coarse, and a 12288^2 run is the obvious next step (Section 8). Fig. 3 plots the table.



figures_v2.json · conv-20261021/analysis/convergence_summary.csv

6.2 Softening systematic

The `eps2x` variant is the one failed convergence test. Doubling ε from 0.08 d to 0.16 d lowers the CDM subhalo count per Milky Way analogue by 9 per cent, raises the WDM count by 6 per cent, raises the WDM/CDM satellite ratio by 20 per cent, and moves M_50 to 0.80 ± 0.08 of the base value, i.e. M_50/M_hm ≈ 0.30 instead of 0.38. The FoF halo counts and P(k) are unchanged at the per cent level, so the effect is confined to the inner structure of haloes, where the softening sets how small a subhalo survives and how much mass the unbinding keeps. The direction (the smallest CDM subhaloes are the first to be lost when the force is softer) is the expected one, and it is the two-dimensional face of the softening-driven artificial disruption discussed by van den Bosch & Ogiya (2018). The variant also takes 1250 instead of 1720 steps (Section 2.2), but `eta_half` shows the step size is not the cause. We attach a ∼20 per cent softening systematic to the two-dimensional M_50; a run with ε = 0.04 d would show whether the base value is itself converged from above. It was not run.

6.3 Completeness study

The finder was rerun on the CDM snapshots of the *original* `exp-20261021-0022` run (the pre-watchdog-fix run of Section 6.7, after the finder-softening fix) with n_min = 40 and 80, with every second particle ID removed (`down2` , particle mass ×2), and with f_esc = 2 and 4. The study gives the mean number of subhaloes per Milky Way analogue in 0.2 dex bins of M_peak for each run. These per-host counts come from the original run and from the study's own host definition, and their totals (66 above 160 m_p for the base run against 38.5 within R_host in Table 2) show that they are not restricted to R_host; only ratios between runs of the study are used. That the study was made on the original run rather than the re-run is a weakness in principle, but Section 6.7 shows the two runs agree in subhalo and FoF statistics within errors, and the completeness criterion is a ratio between two finder passes on the same snapshots, which cancels the run-level differences at first order.

The completeness criterion is: the down-sampled run is converged with the base at every bin with $|N_{down}/N_{base} - 1| \leqslant 0.1 + 2/\sqrt{(30 N_{base})}$. The ratios are 0.69, 0.68, 0.77 and 0.80 in the four bins centred at 50, 79, 126 and 200 m_p and 1.00, 0.91, 0.91, 1.04, 0.96, 0.90, 0.80 from 316 to 5012 m_p, so convergence holds from the bin centred at 316 m_p, and the same particle number at full resolution gives M_complete = 316/2 = 158 m_p, rounded to the 160 m_p used throughout. The fit sample therefore starts at about 160 bound particles at peak; subhaloes at a = 1 can have far fewer, down to n_min = 20.

Table 7. Finder-parameter sensitivity: subhaloes per Milky Way analogue with M_peak ⩾ 158 m_p (sum of the bins from the one centred at 200 m_p upward; the study's host definition, see text). The `down2` row is at half resolution, so its deficit is incompleteness, not a finder effect.

Run	n_min	f_esc	Particles	N(M_peak ⩾ 158 m_p)	Relative to base
base	20	e	all	66.2	1.000
fesc2	20	2	all	65.6	0.990
fesc4	20	4	all	59.8	0.903
nmin40	40	e	all	61.9	0.935
nmin80	80	e	all	50.2	0.758
down2	20	e	half	60.0	0.906

6.4 f_esc and n_min sensitivity

Table 7 shows that lowering the unbinding reference radius from eR_ext to 2R_ext changes the count above M_complete by 1 per cent, and raising it to 4R_ext lowers it by 10 per cent (a larger reference radius makes the reference potential deeper, so marginal particles are classified unbound and marginal subhaloes drop below n_min). The 10 per cent figure is the "count systematic" marginalised in the capped limit (Table 5). The n_min dependence is larger: at n_min = 40 the count above 158 m_p in M_peak falls by 6.5 per cent and at 80 by 24 per cent, because heavily stripped subhaloes whose current bound mass has fallen below n_min are lost even though their peak mass is well resolved. This is why counts are keyed to M_peak with n_min as small as the finder tolerates; it is the same bookkeeping problem as in three dimensions, but the amount of mass loss, in a logarithmic potential with a reference-radius unbinding, is not the three-dimensional amount.

6.5 Fragmentation

The M_lim values of Section 3.5 are 1.7-37 times below M_complete, and the shape-based cleaning flags no subhaloes in the two lighter models and 27 (20 inside host radii) in the heaviest. The fraction of resolved subhaloes below the shape cut is at most 0.4 per cent (CDM itself). Fragmentation therefore does not affect the fits; it would matter if M_complete were lowered towards n_min, which is the regime where the M_hm = 3000 model's M_lim = 93 m_p lies. The analysis can be rerun with the shape cut applied at any mass or with no cleaning; those variants exist for the original experiment but were not rerun on the corrected one.

6.6 Reproducibility and determinism

Two sources of run-to-run variation were separated. First, GPU atomics and thread scheduling: the chaining-mesh counting sort assigns in-cell positions with an atomic counter, whose order is scheduling dependent, so without further care the PP pair sum would be evaluated in a different order each run and the forces would differ at the rounding level. A rank pass that sorts particles within each cell by a stable key before the final scatter fixed this; the test ran the full 6144^2 system on the 4096^2 mesh for 100 fixed steps of $\Delta \ln a = 0.01$ twice and found 0 of 37 748 736 particles differing in position or velocity. Second, the adaptive step: A_max is reduced on the GPU and read back asynchronously, so the CFL step planned on the CPU uses "the latest available maximum", which may lag by a scheduling-dependent number of steps. Two science runs can therefore take slightly different step sequences and are not bit-identical even with

deterministic forces. This is not fixed. Its size is the `control_repeat` row of Table 6: P(k) identical to 3×10^{-5} in the worst bin, FoF counts identical to 0.1 per cent, subhalo counts per Milky Way analogue reproduced to 1.4 ± 1.6 per cent, and 99.8 per cent median shared particle fraction between matched hosts. The reproducibility floor for subhalo counts is thus 1-2 per cent, far below the softening and cosmic-variance effects.

6.7 Numerical history

Four changes during the project affected the data and are recorded here.

(i) The original `cdm` and `wdm_khm0.31` runs of `exp-20261021-0022` were made before the PP-force submission was split into bounded chunks to avoid the Windows GPU watchdog (Section 2.3). Those two runs showed a mean velocity drift of order 10^{-2} (against an rms of 220) and excess low-k growth: their low-k P/P_lin had drifted to 1.093 (CDM) and 1.126 (WDM) by a = 0.053, where the re-runs give 0.998 and 0.992. They were re-run with the current code as the `control` variant of the convergence suite and those re-runs are the ones used here. The other two models of the original experiment were checked (mean velocity $\sim10^{-5}$, particle-level agreement with the control CDM run at a = 0.1) and kept. The re-runs were compared with the originals with the convergence tool, which treats the original as a variant of the re-run; the ratios (original-run comparison measured with the convergence tool against the original runs; not written to a file) are in Table 8. Every subhalo and FoF statistic agrees within its error, the P(k) medians agree to 2 per cent, and the only large deviation is the worst low-k bin of the WDM run (0.686), which is the low-k drift itself. The completeness study of Section 6.3 therefore stands.

Table 8. Agreement of the clean re-runs (`conv-20261021/control`) with the original `exp-20261021-0022` runs, as ratios re-run/original, measured with the convergence tool against the original runs.

Quantity	CDM	WDM (M_hm = 3000)
N_FoF(> M_complete)	1.001 ± 0.013	1.023 ± 0.023
N_FoF(> 100 M_complete)	0.94 ± 0.14	0.91 ± 0.14
Subhaloes per Milky Way host	1.067 ± 0.053	0.99 ± 0.19
WDM/CDM satellite-count ratio	0.92 ± 0.15	
M_50 (β, γ fixed)	1.15 ± 0.20	
P(k) median, k ⩽ half Nyquist	1.002	0.982

Quantity	CDM	WDM (M_hm = 3000)
Worst low-k bin		0.686

(ii) The finder initially used a default softening of 0.75 PM cells (0.366) in the unbinding potential instead of the simulation's PP softening (0.026), a factor 14.06 too large; the first analysis found 23 140 CDM subhaloes at a = 1. With the correct softening the count is 84 841. All results in this paper use the corrected finder.

(iii) Host matching was positional in the first analysis (nearest host within 0.5R_host and a factor 3 in mass) and is now by particle IDs (Section 3.4); the positional code remains as a fallback.

(iv) The determinism work of Section 6.6.

7 Discussion

7.1 What transfers from two to three dimensions, and what does not

What transfers is the methodology: the SUBFIND candidate-and-unbind logic, tracer-based trees, M_peak bookkeeping, ID-based host matching, pair-level bootstraps, the fragmentation flag, the closure test between a fitted suppression and the direct simulations, and the capped likelihood. Each of these could be lifted into a three-dimensional pipeline unchanged, and the testbed shows which of them matter: softening and n_min at the 10-25 per cent level in counts, f_esc at 1-10 per cent, reproducibility at 1-2 per cent, cosmic variance at 10-25 per cent for 30 hosts. The sizes of these effects are two-dimensional numbers; the ranking and the way to measure them are what carries over.

What does not transfer is any amplitude. The half-suppression mass in units of M_hm is 0.40 here against 2.5-2.7 in three dimensions; the functional shape is much steeper (f(M_hm) = 0.97 against 0.27-0.31); and re-expressing mass through σ(M) gives x_50 = 1.135 against 0.92-0.93. Several differences contribute and we have not separated them: (a) the definition of M_hm as a disc of diameter λ_hm is a convention, and a different geometric factor would shift M_50/M_hm by a factor of order 4 but not the shape; (b) for n = −1.5 the two-dimensional σ(R) ∝ R^(−(n+2)/2) gives σ(M) ∝ M^(−(n+2)/4) = M^(−1/8), so flat that a cutoff in P(k) appears in σ(M) only well below the half-mode scale, which is the direction of the observed shift; (c) two-dimensional haloes have logarithmic potentials, no escape, flat rotation curves and a different tidal field, so the mass loss and

survival of subhaloes, and the orbital decay that in three dimensions is called dynamical friction, are quantitatively different processes that we have not characterised; (d) the mass-function shape itself is not Press-Schechter in two dimensions (Section 4.3). The honest summary is that the two-dimensional model universe is a different universe with the same algorithms.

7.2 Comparison with the three-dimensional fits

The three-dimensional references used are the Lovell et al. (2014) genuine-subhalo fit (1 + 2.7M_hm/M)^(−0.99) (applied to M_peak by Nadler et al. 2020, 2021a), the COZMIC I M_peak fit [1 + (2.5M_hm/M_peak)^0.9]^(−1), and the Lovell (2020) bound-mass and central fits. In all of them the suppression is a gentle power law in M_hm/M with $\beta \approx 1$ and $\gamma \approx -1$, reaching 50 per cent at 2.3-2.7M_hm (1.06 for bound mass). The two-dimensional fit has $\beta \approx 8$ and $\gamma \approx -0.1$: a near step. We checked that this is not a fitting artefact by (i) the per-bin ratios (Fig. 1), which are consistent with unity (or above it) down to M_peak/M_hm ≈ 1 and fall steeply below ≈ 0.8, (ii) the single-model fixed-shape M_50 of the convergence suite (0.38 for M_hm = 3000 alone), and (iii) the central-halo fit, which gives M_50/M_hm = 0.36 with a very different (β, γ) pair, as expected for a near-step function in a degenerate parametrisation.

7.3 What the testbed is good for

It is good for anything where the question is "does this algorithmic choice change the answer, and by how much", at a cost of 12-35 minutes per variant for simulation plus 4 minutes for finding and 14 s for analysis on one desktop. Examples carried out here: the completeness criterion from down-sampling, the n_min and f_esc dependence, the fragmentation flag rate, the ID-matching fraction as a function of M_hm (88, 78 and 46 per cent), the reproducibility floor, and the closure test that exposed a failure of the three-parameter suppression form when WDM counts exceed CDM near M_hm. Examples it is not good for: anything quoted in physical units of a three-dimensional cosmology; the normalisation of satellite luminosity functions; constraints on M_hm or on the WDM particle mass.

7.4 Questions a subhalo expert would ask

Is a logarithmic potential a sensible place to run SUBFIND? The candidate step is dimension-agnostic; only the unbinding needs a reference level, and there is no physically preferred one. We chose f_esc = e by matching the three-dimensional escape budget at

the boundary and measured the sensitivity (1 per cent for f_esc = 2, 10 per cent for 4). The choice is a convention and is reported as a systematic.

Why are there more subhaloes than in CDM just above M_hm? In the M_hm = 300 model the paired bin ratios exceed unity by 10-30 per cent in four consecutive bins between 1 and 5 M_hm (1-2σ each); the M_hm = 3000 model hints at the same with few objects; the M_hm = 30 model has that mass range below M_complete. We have not identified the cause. The candidates, roughly in the order we would test them, are:

1. *Host-matching selection.* The paired WDM hosts are more massive than their CDM partners (median 1.05 for M_hm = 300, 1.14 for 3000; Section 3.4) because mass that in CDM is in small neighbouring haloes is in WDM part of the host. If subhalo number scales roughly with host mass this accounts for about 5 of the 9 per cent per-host excess of the M_hm = 300 model, but not obviously for 10-30 per cent in individual bins. A test is to recompute the bin ratios at matched *WDM* host mass, or to normalise counts by host mass.
2. *Later formation and different concentrations.* Haloes near the cutoff form later in WDM and are less concentrated; in a logarithmic potential with a saddle-point boundary this changes the candidate extent R_ext, hence R_ref, hence how much mass the unbinding keeps, and it changes M_peak (less early stripping, so a higher peak for the same object). This is a finder-plus-dynamics effect that would be visible as a systematic difference in M_peak/M_bound or R_ext for matched subhaloes.
3. *Artificial fragmentation below the resolution limit.* M_lim/M_hm = 0.1 for this model and the shape cut flags nothing above 0.5M_lim, so fragments at 1-5 M_hm would have to be spurious objects that the Lovell et al. (2014) criterion, transcribed rather than calibrated in two dimensions, does not catch. A direct test is the Lagrangian shape and filament association of the excess objects.
4. *Cosmic variance at fixed phases.* The three WDM models share one seed. `seed2` shows that 30 hosts carry a 25 per cent scatter in counts, but a same-seed ratio removes most of it; what remains is the realisation-specific response of this particular set of hosts to the cutoff. A second seed for the M_hm = 300 model is the cheapest decisive test (16 minutes).
5. *Genuine two-dimensional dynamics.* Fewer small-scale encounters in a smoother WDM host, or a different tidal field, could let subhaloes just above the cutoff survive longer or lose less mass. This would be the interesting answer and the last one to accept.

What would separate them: a second seed for the WDM models (4); a `np12288` or `down2` comparison of the excess bins specifically (3, and whether it moves with resolution); matched-subhalo comparisons of M_peak, bound mass and R_ext between CDM and WDM (2); and ratios at fixed WDM host mass (1). None of these was done. Until they are, the excess is a feature of this pipeline in this universe, and the capped likelihood exists precisely so that it cannot tighten a limit.

Is M_peak reliable with 16 snapshots? Partially. The spacing is 0.153 in ln a; a subhalo that peaks between snapshots has its M_peak underestimated, and 7 per cent of searched links are lost. Both effects are present in all models and partly cancel in ratios, but they are not corrected and they are not identical between models if WDM subhaloes peak later.

Is the pairing selection biasing the M_hm = 3000 result? Only 46 per cent of CDM hosts and 24 of the 30 analogues have counterparts; the six unmatched analogues are those whose Lagrangian region is reorganised by the cutoff. The ratios are computed on pairs only, so they describe the matched population, which is more massive in WDM than in CDM by a median 14 per cent.

How complete is the sample at 160 particles? The down-sampling test says counts above 158 m_p in M_peak are converged at the 10 per cent plus 2σ level; the bins immediately below are 20-30 per cent low. Subhaloes near the cut have a = 1 bound masses that can be far smaller than 160 particles, which is why n_min matters so much.

Is the softening adequate? ε = 0.08 d is conventional, but doubling it changes M_50 by 20 per cent. A halved-softening run is needed to show convergence from the other side; it was not run.

Is the 2D halo mass function understood? No. Press-Schechter with the two-dimensional top-hat δ_c = 1.466 does not describe the FoF counts; a Sheth-Tormen form fits only with free (p, q). We use the mass function only as a check of hierarchical growth.

What about the in-box normalisation? The target σ = 1.15 for 10^6-particle hosts is not realised in the box (0.84 is), so the analogues are rarer and slightly smaller than designed; this affects host statistics, not the methods, and was left as is so that the four models differ only in T(k).

8 Conclusions and future work

We have built and validated a two-dimensional comoving P³M code, a two-dimensional SUBFIND with merger trees, and an analysis pipeline, and used them to run a complete CDM versus WDM subhalo experiment with a convergence suite and finder studies on one desktop in a few hours of compute. The code reproduces D = a to 0.3 per cent in the linear regime, agrees with an independent power-spectrum measurement to 0.05 per cent, and is deterministic in its forces. The WDM suppression of the subhalo M_peak function in this universe reaches 50 per cent at M_50/M_hm = 0.398 [0.385, 0.416], a factor 6-7 below three-dimensional results (part of it the M_hm convention), and the difference survives a change of variable to σ(M); the amplitude does not transfer. All convergence tests pass at the 10 per cent or 2σ level except M_50 under doubled softening (0.80 ± 0.08) and single P(k) bins near the mesh scale and in the fundamental mode of a second seed; the particle-number test is coarse and marginal for M_50 and for FoF counts at the completeness limit. The M_hm = 300 model has an unexplained 10-30 per cent excess of subhaloes over CDM at 1-5 M_hm; the emulator built from the fitted suppression closes for the other two models and fails by 12 per cent for this one because the fit form cannot represent an excess. The mock limit, M_hm < 479 m_p, is a closure exercise on a CDM mock in a toy universe and says nothing about dark matter.

Future work, in order of value: a second seed for the WDM models and a resolution test of the excess bins, to decide whether the excess is selection, resolution, variance or dynamics; a halved-softening run and a 12288² run to bracket the softening and resolution dependence of M_50; more snapshots (or on-the-fly M_peak tracking) to remove the M_peak bias; a suppression form with an additional parameter (or a non-parametric emulator) so that closure can be met; and, once the finder and matching are trusted, the same experiment in three dimensions with the same code structure, where the amplitudes would mean something.

Data and code availability

All runs and outputs are on the author's workstation; the code is not yet public.

- Fiducial experiment: `D:\cosmo-runs\experiments\exp-corrected\` (directory junctions to `conv-20261021\control\{cdm,wdm_khm0.31}` and `exp-20261021-0022\{wdm_khm0.99,wdm_khm3.12}`; see `README.txt`). Analysis outputs in `exp-corrected\analysis\`: `experiment_summary.json`, `shmf.csv`,

`suppression_fit.csv`, `suppression_sigma.csv`, `closure.csv`, `likelihood.csv`, `radial.csv`, `satellites_lf.csv`, `hosts.csv`, `pairing.csv`, and the printed summary `D:\cargo-target\tmp\cosmo_analysis_corrected.txt`.

- Run metadata: `meta.json` and `run.log` in each model directory; finder metadata in `subhalos\subfind_meta.json`.
- Convergence suite: `D:\cosmo-runs\experiments\conv-20261021\` with `analysis\convergence_summary.csv`, `convergence.csv`, `convergence_meta.json`. The re-run-versus-original comparison of Table 8 was produced with the same tool but not saved.
- Completeness and finder studies: `D:\cosmo-runs\experiments\exp-20261021-0022\completeness.csv` and `subfind_studies\`.
- Validation runs: `D:\cosmo-runs\20261020-233820\` (`analysis\summary.json`), paired-phase growth runs `D:\cosmo-runs\20261020-233221\` and `20261020-233559\` (`pk.csv`); determinism test `D:\cargo-target\tmp\dt1.txt`.
- Code: `C:\Users\ricej\OneDrive\Documents\RustGames\gravity-sandbox\` (simulation: `src\cosmo.rs`, `src\pm.rs`, `src\shaders\cosmo.wgsl`), `subhalo-finder\` (`src\units.rs`, `src\subfind.rs`, `src\merger.rs`), `cosmo-analysis\` (`src\wdm.rs`, `src\experiment.rs`, `src\galaxy.rs`, `src\hmf.rs`, `src\convergence.rs`). All Rust; the GPU code is WGSL via wgpu.
- Figure data: `D:\cargo-target\tmp\paper\figures_v2.json`.

Acknowledgements

The simulation, finder and analysis code were written with AI assistance (Claude) and run on a desktop with an NVIDIA GeForce RTX 5080 and an AMD Ryzen 7 9850X3D. This paper was drafted and revised with AI assistance from the project's output files and source code; every number should be checked independently against those files before the draft is used.

References

- Angulo R. E., Pontzen A., 2016, MNRAS, 462, L1
- Bagla J. S., Engineer S., Padmanabhan T., 1998, ApJ, 495, 25
- Banik N., Bovy J., Bertone G., Erkal D., de Boer T. J. L., 2021, JCAP, 2021(10), 043 (arXiv:1911.02663)

- Beacom J. F., Dominik K. G., Melott A. L., Perkins S. P., Shandarin S. F., 1991, ApJ, 372, 351
- Davis M., Efstathiou G., Frenk C. S., White S. D. M., 1985, ApJ, 292, 371
- Enzi W., et al., 2021, MNRAS, 506, 5848
- Gilman D., Birrer S., Nierenberg A., Treu T., Du X., Benson A., 2020, MNRAS, 491, 6077
- Hockney R. W., Eastwood J. W., 1988, Computer Simulation Using Particles. Adam Hilger, Bristol
- Lovell M. R., Frenk C. S., Eke V. R., Jenkins A., Gao L., Theuns T., 2014, MNRAS, 439, 300 (arXiv:1308.1399)
- Lovell M. R., 2020, ApJ, 897, 147 (arXiv:2003.01125)
- Melott A. L., Shandarin S. F., 1989, ApJ, 343, 26
- Nadler E. O., et al., 2020, ApJ, 893, 48
- Nadler E. O., et al., 2021a, Phys. Rev. Lett., 126, 091101 (arXiv:2008.00022)
- Nadler E. O., Birrer S., Gilman D., Wechsler R. H., Du X., Benson A., Nierenberg A. M., Treu T., 2021b, ApJ, 917, 7 (arXiv:2101.07810)
- Nadler E. O., An R., Gluscevic V., Benson A., Du X., 2025, ApJ, 986, 127 (COZMIC I; arXiv:2410.03635)
- Onions J., et al., 2012, MNRAS, 423, 1200
- Press W. H., Schechter P., 1974, ApJ, 187, 425
- Quinn T., Katz N., Stadel J., Lake G., 1997, arXiv:astro-ph/9710043
- Schneider A., Smith R. E., Macciò A. V., Moore B., 2012, MNRAS, 424, 684 (arXiv:1112.0330)
- Sheth R. K., Tormen G., 1999, MNRAS, 308, 119
- Springel V., White S. D. M., Tormen G., Kauffmann G., 2001, MNRAS, 328, 726
- Stücker J., Angulo R. E., Hahn O., White S. D. M., 2022, MNRAS, 509, 1703 (arXiv:2109.09760)
- van den Bosch F. C., Ogiya G., 2018, MNRAS, 475, 4066
- Viel M., Lesgourgues J., Haehnelt M. G., Matarrese S., Riotto A., 2005, Phys. Rev. D, 71, 063534
- Wang J., White S. D. M., 2007, MNRAS, 380, 93
- Zel'dovich Ya. B., 1970, A&A, 5, 84